

STUDENT ACTIVITY FILE

INNOVATIVE DESIGN LAB (IDL) - I

Academic year: 2026-2027 | Department: Computer Science & Engineering

Field	Details
Academic Year	2026-2027
Department & Section	Computer Science and Engineering Section: A & C
Innovation Ecosystem Project ID	SAIRAM-IDL1-2026-CSE-09
Title of the Project	SENTINEL: Autonomous Hybrid-AI SOC Analyst & Investigative Engine
Inter disciplinary Project	☒ Yes ☐ No Team
Members (Max. 3)	1. Gokula Kannan M — SEC25CS196 2. Lakshan M — SEC25CS036 3. Sivabalan T — SEC25CS101
Project Version	☒ Version 1 – New Project ☐ Version 2 – Pass out Student Project
Domain Name	Cybersecurity, Artificial Intelligence & Autonomous Systems
IEEE Society & Community	IEEE Computer Society IEEE Student Branch Sairam
Club and Cells	Sairam Innovation Ecosystem / Cyber Security Club
Name of Department IDL - I Coordinator	Dr. A. SHEELA Faculty ID: IDL-FAC-01
Name of the Supervisor	Dr. A. SHEELA Faculty ID: IDL-FAC-02
Name of External Guide	Senior Tech Leads & Cyber Forensic Specialists
Designation & Organization	Cyberdome Investigations / Zoho Corporation

SUSTAINABLE DEVELOPMENT GOALS (SDG) MAPPING

SDG	Goal Number With Name	SAP Code with Explanation
Primary SDG:	SDG 9: Industry, Innovation, and Infrastructure	1. SAP-SDG9-CYBER-01: Resilient cyber infrastructure & AI SOC automation. 2. SAP-SDG9-CYBER-02: Zero-trust cloud-edge security architecture. 3. SAP-SDG9-CYBER-03: Scalable threat intelligence platform.
Secondary SDG:	SDG 16: Peace, Justice, and Strong Institutions	1. SAP-SDG16-SEC-01: Evidence privacy & chain-of-custody enforcement. 2. SAP-SDG16-SEC-02: Citizen PII protection in digital investigation. 3. SAP-SDG16-SEC-03: Transparent forensic audit logging.
Tertiary SDG:	SDG 8: Decent Work and Economic Growth	1. SAP-SDG8-AUTO-01: Reduction of SOC analyst alert fatigue. 2. SAP-SDG8-AUTO-02: Decreasing MTTR from 45 mins to < 30 seconds. 3. SAP-SDG8-AUTO-03: Workforce efficiency & burnout prevention.

ACTIVITY LOG INDEX TABLE (WEEKS 1 - 13)

S.N o.	Title	Start Date	Completion Date	Mark (Out of 10)	Remarks	Signature of IDL - I Coordinator
1	Orientation & Entrepreneurial Mindset	01/08/2026	07/08/2026	10	Completed	
2	Problem Identification & Passion CV	08/08/2026	14/08/2026	10	Completed	
3	Customer Segmentation & Persona Creation & JTBD	15/08/2026	21/08/2026		In Progress	
4	Ideation, Market Size & Competitor Analysis	22/08/2026	28/08/2026		Upcoming	
5	Milestone1:Problem-Solution Fit Presentation	29/08/2026	04/09/2026		Upcoming	

OUTSIDE WORLD PROJECTION

Details of the Competition Attended (Hackathon / Idea Pitching)

S.No	Name	Date	Place	Remark	Signature
1	Hac'KP 2026 (7th Edition National Hackathon by Kerala Police Cyberdome)	14/08/2026	Zoho Corporation, Chennai	Selected for National In-Person Pitching Round	
2					

Design Patent / Idea Patent Publication / Conference Paper

S.No	Title / Forum	Date	Place / Journal	Remark	Signature
1	Confidentiality-Preserving Autonomous SOC Triage via Tokenized Edge-Cloud Multi-Agent Systems	August 2026	IEEE / USENIX Track	Paper Manuscript & Patent Drafted	
2					

IDL – I Coordinator**Creative Innovator****H.O.D**

Dr. A. SHEELA

Gokula Kannan M (Team Lead)

Head of Department

WEEK – 1

1. Orientation & Entrepreneurial Mindset

• Understand entrepreneurship & innovation

Security Operations Centers (SOCs) generate over 10,000 security logs daily, creating severe alert fatigue where human analysts spend 30–45 minutes manually triaging single events. Commercial AI endpoints cannot be deployed directly due to data privacy regulations and risks of uploading confidential network payloads to external servers.

The proposed solution, SENTINEL, introduces Reversible Tokenized Pseudonymization, scrubbing 100% of sensitive PII, IP addresses, and hostnames locally before AI reasoning occurs. This enables enterprise-grade autonomous triage at \$0 operational cost for 90% of routine alerts using local consumer GPU hardware (NVIDIA RTX 3050).

• Opportunity recognition, entrepreneurial traits and growth mindset

The global SIEM automation and threat intelligence market is valued at \$13.2 Billion+ with an 18.5% CAGR. Primary target markets include Enterprise SOCs, Managed Security Service Providers (MSSPs), Air-Gapped Defense Networks, Cyber Crime Units, and Startups. Technical resourcefulness is demonstrated by quantizing 70B parameter models down to 4-bit (IQ3_M) GGUF format to execute high-capacity intelligence on edge workstations.

• Startup journey from idea to venture

1. Problem Formulation: Identifying SOC alert fatigue and data privacy constraints.
2. Proof-of-Concept: Engineering local Zero-Trust Data Sanitizer (src/sanitizer.py).
3. MVP Architecture: Building 3-Tier Hybrid AI Router (src/router.py).
4. Validation & Pitching: Selected for National Pitching Round at Hac'KP 2026 @ Zoho Corporation.
5. Commercialization: Deploying open-source SaaS framework and enterprise MSSP engine.

WEEK – 2

2. Problem Identification & Passion CV

- **Identify real-world problems based on personal interests**

Modern digital investigations face a severe bottleneck where human triage speed cannot match machine-speed cyber attacks. Public commercial cloud LLMs violate legal data privacy and chain-of-custody mandates when processing un-sanitized evidence.

- **Skills, experiences, and societal needs**

Small businesses, hospitals, and educational institutions are unable to afford \$100k+/year enterprise SIEM subscriptions (e.g., Splunk or Cortex XSOAR), leaving critical infrastructure vulnerable to ransomware. SENTINEL addresses this societal gap by delivering a self-hosted, privacy-compliant, autonomous AI co-pilot operating on standard hardware.

- **Prepare a Passion CV to identify potential entrepreneurial opportunities**

Team Lead: Gokula Kannan M (SEC25CS196 | CSE-A)

- Domain: Software Architecture, Systems Engineering, Project Governance & Venture Strategy.
- Project Contribution: Overall system architecture, sprint management, and leading the national pitch at Hac'KP 2026 @ Zoho Corporation.

Team Member 1: Lakshan M (SEC25CS036 | CSE-A)

- Domain: Full-Stack Web Development, React.js UI, WebSockets Integrations.
- Project Contribution: Real-time incident dashboard engineering and Human-in-the-Loop (HITL) action approval interface.

Team Member 2: Sivabalan T (SEC25CS101 | CSE-C)

- Domain: AI Engine Architecture, Zero-Trust Privacy Engineering, Quantization & Vector RAG.
- Project Contribution: Development of Zero-Trust Data Sanitizer (src/sanitizer.py) and 3-Tier AI Router (src/router.py).